H.T.No: | | | | | | | | | | | Course Code: **241MA002**

ADITYA UNIVERSITY

B. Tech – II Semester End Examinations Supplementary – SEP 2025

DIFFERENTIAL EQUATIONS & VECTOR CALCULUS
(Common to CE, EEE, ME, ECE, CSE, IT, Min.E, PT, AIML & CSE(DS))

Time: 3 hours **Max. Marks: 100**

Answer ONE question from each unit
All Questions Carry Equal Marks
All parts of the questions must be answered at one place only

UNIT-I

1	a	Solve the differential equation $y(1+xy)dx+x(1-xy)dy=0$.	L2	CO1	[10M]
	b	A generator battery ofemf 100 volts **is** connected in series with a **resistor** of 10 ohms and an inductor of 2 Henries. If the switch is closed at time t=0, find the currentI(t) in the circuit for t>0.	L3	CO1	[10M]

(OR)

2	a	Solve the differential equation $\frac{dy}{dx}+xy=xy^2$.	L2	CO1	[10M]
	b	Find the orthogonal trajectories of the family of circles $x^2+y^2=r^2$.	L3	CO1	[10M]

UNIT-II

3	a	Solve the differential equation $(D^2+3D+2)y=x^2,\ D\equiv\frac{d}{dx}$.	L2	CO2	[10M]
	b	Solve the differential equation $\frac{d^2y}{dx^2}-y=\frac{2}{1+e^x}$ using the method of variation of parameters	L3	CO2	[10M]

(OR)

4	a	Solve the differential equation $\frac{d^2x}{dt^2}+3\frac{dx}{dt}+2x=\cos t$.	L2	CO2	[10M]
	b	An LCR circuit connected in series has $R=3$ ohms, $C=0.5$ farad, $L=1$ henry and an applied voltage $E=10$ volts. Assuming no initial current and no initial charge at $t=0$ when the voltage is first applied, find the subsequent current in the system at any time t.	L3	CO2	[10M]

UNIT-III

5	a	Solve the partial differentia equation $xp-yq=y^2-x^2$, where $p=\frac{\partial z}{\partial x}$ and $q=\frac{\partial z}{\partial y}$.	L3	CO3	[10M]
	b	Solve $\frac{\partial^3 z}{\partial x^3}-5\frac{\partial^3 z}{\partial x^2\partial y}+8\frac{\partial^3 z}{\partial x\partial y^2}-4\frac{\partial^3 z}{\partial y^3}=e^{2x+y}$.	L2	CO3	[10M]

(P.T.O)

(OR)

6	a	Solve the partial differentia equation $y^2zp + x^2zq = y^2x$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.	L3	CO3	[10M]
	b	Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = e^{3x+y}$.	L2	CO3	[10M]

UNIT-IV

7	a	Find the directional derivative of the function $f = xy + yz + zx$ in the direction of the vector $\bar{i} + 2\bar{j} + 2\bar{k}$ at the point $(1, 2, 0)$.	L2	CO4	[10M]
	b	Show that the vector field $\vec{f} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is irrotational and hence find the scalar potential of $\vec{f}$.	L3	CO4	[10M]

(OR)

8	a	Find the div (grad ϕ) where $\phi = x^2 - y^2 + 2z^2$.	L2	CO4	[10M]
	b	Find the constants a, b, c such that the vector field $\vec{f} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational and hencefind its scalar potential of $\vec{f}$.	L3	CO4	[10M]

UNIT-V

9	a	If $\vec{f} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$, evaluate $\int_V curl\ \vec{f}\ dV$, where V is the volume of the region bounded by the planes $x = 0,\ y = 0, z = 0$ and $2x + 2y + z = 4$.	L2	CO5	[10M]
	b	Verify Green's Theorem for $\int_C (x^2 - xy^3)\,dx + (y^2 - 2xy)\,dy$ where C is the square with vertices (0, 0), (2, 0), (2, 2) and (0, 2).	L3	CO5	[10M]

(OR)

10	a	Find the work done in moving a particle in the force field $\vec{f} = y\vec{i} + 3x\vec{j}$ along the upper part of the circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$.	L2	CO5	[10M]
	b	Using Gauss divergence theorem, evaluate $\iint_S \bar{F}.\bar{n}\,dS$ where $\bar{F} = 2xy\,\bar{i} + yz^2\,\bar{j} + xz\,\bar{k}$ and S is the surface of parallelepiped $0 \le x \le 2, 0 \le y \le 1, 0 \le z \le 3$.	L3	CO5	[10M]

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